

The lifetime of a record is the thermalization time: a measured, derived, and falsifiable horizon law for the memory arrow of time

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In the standard account of the arrow of time, the direction in which memories accumulate is slaved to the thermodynamic gradient: records form on the entropy-increasing side of a low-entropy boundary condition. That statement is usually qualitative. Here we make it quantitative. In an exactly bit-reversible lattice gas we place two clocks on the same dynamics: the entropy-saturation time t_S (coarse-grained entropy reaches $0.9 S_{\max}$) and the *record horizon* t^* , the time at which the mutual information between a planted macroscopic fact and an optimal readout of the present coarse state falls below half a bit. Across scrambling rates and system sizes we find $t^* = \kappa t_S$ with κ of order one and flat ($\kappa = 1.08$, $R^2 = 0.82$ over a $7\times$ range; size-robust, $\bar{\kappa} = 0.99$ for $L = 48 \dots 128$): *the readable record dies exactly when entropy saturates*. The same law, with substrate-dependent order-one κ , holds in a continuous hard-disk gas ($\kappa = 1.00$) and in exactly reversible Clifford circuits of up to 128 qubits ($\kappa = 0.79$), and survives a Haar-random (non-Clifford) control. The law is derived, not merely fitted: entropy deficit and record signal relax through the same slowest mode, making κ a ratio of logarithms of amplitudes and thresholds; the resulting parameter-free postdiction reproduces the two crossing times to 5% and 1% median error. The law is falsifiable, and fails exactly where it must: a conservation law protecting the record decouples t^* from t_S . Finally, records are stored redundantly (classical Darwinism): redundancy grows as N^α with a conservative $\alpha_\infty = 0.92 \pm 0.04$ classically and $\alpha = 1$ exactly for quantum broadcast states. These results quantify the sense in which the experienced past is the direction of the entropy gradient—while leaving the low-entropy boundary itself unexplained.

I. INTRODUCTION

Microscopic dynamics—classical Hamiltonian flow, unitary quantum evolution, or the exactly bit-reversible cellular automata used below—has no arrow of time. The observed asymmetry is standardly traced to a low-entropy boundary condition (the Past Hypothesis) together with coarse-graining [1–5]. Within this picture the *memory* (or “psychological”) arrow is derivative: a physical record is a present macrostate correlated with a macrostate at another time, and such correlations are abundant only on the entropy-increasing side of the boundary [6–11]. Reichenbach’s fork asymmetry, Wolpert’s analysis of memory systems, and Mlodinow and Brun’s argument that a robust record-forming system must sit on the entropy-increasing side all support the same conclusion: *memory points down the entropy gradient*.

These arguments are, however, largely qualitative. They establish which *direction* records point, but not the quantitative structure of record-keeping: How long does a record remain readable? What sets that lifetime? How is the record physically stored, and how does its accessibility scale with the size of the environment that stores it? The quantum Darwinism program [12–15] answered the storage question for decohering quantum systems—objective, classical facts are those redundantly imprinted in many environment fragments—and Riedel, Zurek, and Zwolak studied the rise and fall of that redundancy in time [16]. Observational entropy [17–20] supplies the information-theoretic bookkeeping for coarse observers of closed systems. What has been missing is a direct, con-

trolled measurement, in exactly reversible dynamics, of the *lifetime* of a record and its relation to the thermodynamic clock of the same system.

This paper reports such measurements, together with a mechanism that explains them and a regime in which they demonstrably fail. Our testbed is deliberately minimal: exactly reversible model dynamics (a Margolus block cellular automaton [21, 22], an event-driven hard-disk gas, and Clifford circuits [23, 24]), a coarse-grained Boltzmann entropy, and a one-bit macroscopic fact planted at the low-entropy boundary whose readability we track with an explicit decoder. Our contributions:

1. **An exact ledger** (Sec. II): over an ensemble of microstates evolved by a bijective map, the fine-grained (Gibbs/Shannon) entropy is conserved to machine precision while the observational entropy rises; the rise is *identically* the growth of information hidden from the coarse observer, $\Delta S_{\text{obs}} = \Delta I$. The record’s decay is the same bookkeeping seen from the fact’s side.
2. **A horizon law** (Sec. III): the record horizon tracks the entropy-saturation time, $t^* = \kappa t_S$ with κ order one, flat across scrambling rates ($7\times$ in t_S) and system sizes.
3. **A mechanism** (Sec. IV): both clocks read the same slowest relaxation mode; κ is a computable ratio of logarithms. A parameter-free postdiction from the fitted mode reproduces t^* and t_S to 5% and 1% median error, and explains why κ is flat in system size, substrate-dependent, and only logarithmically sensitive to threshold conventions.

4. **Universality** (Sec. V): the law reproduces in a continuous hard-disk gas and in reversible Clifford circuits (up to $N = 128$ qubits), and survives a Haar-random statevector control—it is not a lattice or stabilizer artifact.
5. **A falsification boundary** (Sec. VI): the law is contingent on ergodicity. A conservation law protecting the record makes t^* diverge while t_S stays finite. The horizon law is a law, not a definition.
6. **Redundancy scaling** (Sec. VII): near the boundary the record is stored in $R \approx 8$ –10 independent copies; R grows with the number of environment fragments as N^α with a conservative $\alpha_\infty = 0.92 \pm 0.04$ (equally-well-fitting extrapolations reach 0.95–1.00), while a quantum broadcast realizes ideal Darwinism, $\alpha = 1$ exactly—isolating the classical shortfall as a signal-to-noise artifact rather than a limit of the principle.

Throughout we are explicit about what is *not* shown (Sec. VIII): the low-entropy boundary is an input, as it must be for any time-symmetric dynamics [25]; the numerical value of κ is convention-dependent (the law is the proportionality and the collapse); and the classical redundancy exponent is supported, not certified, at ideal value. All code, data, and per-experiment numerical verdicts are public [26].

II. SETUP: SUBSTRATES, RECORDS, AND THE LEDGER

A. Reversible substrates and coarse entropy

Our primary substrate is a Margolus block cellular automaton [21, 22]: a two-dimensional lattice gas on an $L \times L$ grid of binary site occupations, updated in 2×2 blocks by bijective lookup tables with alternating block offsets. A quenched random fraction (“scatter”) of blocks applies a rotation rule, breaking spurious conservation laws so the gas thermalizes cleanly; the map remains exactly bijective, and the inverse map recovers the initial bitstring exactly (verified bit-for-bit [26]). Appendix A gives details. Two companion substrates are used for universality tests: an event-driven hard-disk gas (elastic disks, specular walls; exact between-collision flight) and Clifford brickwork circuits simulated in the stabilizer formalism (Appendix C).

The coarse-grained (Boltzmann) entropy is defined by partitioning the grid into $b \times b$ cells and counting microstates compatible with the cell occupancies $\{n_i\}$:

$$S = \sum_i \ln \binom{b^2}{n_i}, \quad (1)$$

with $S_{\max}$ its microcanonical maximum at the given particle number. Initial states are low-entropy “blobs”:

all particles confined to a region of width w at density ρ , with microstates drawn microcanonically within the blob.

B. Records and the two clocks

A *record* is operationalized as follows. A one-bit macroscopic fact F (a marker of identical particle number placed on the left or the right side of the blob—so the two classes differ in no conserved quantity) is planted at $t = 0$. For each class we evolve an ensemble of K microstates through the same quenched disorder. The readout at time t is a decoder $\hat{F}$ trained on the coarse occupancy vectors of half of the worlds and evaluated on the held-out half; the accessible record is the mutual information $I(F; \hat{F})$ estimated from the held-out confusion matrix (Appendix B). This is a *present-macrostate* readout: the decoder sees only the coarse state at time t , never the history.

The two clocks, defined on the same runs:

$$t_S : \quad \text{first time } S(t) \geq 0.9 S_{\max}, \quad (2)$$

$$t^* : \quad \text{first time } I(F; \hat{F}) < \frac{1}{2} \text{ bit}. \quad (3)$$

The thresholds are conventions; Sec. IV shows the law depends on them only logarithmically, and the time-rescaled collapse plots below make the convention-independent content explicit.

C. The exact ledger

Before measuring lifetimes we fix the bookkeeping. Consider an ensemble of K equiprobable microstates evolved by the (bijective) dynamics. The Shannon entropy H of the microstate distribution is exactly conserved, $H = \ln K$; in our runs ($K = 64$, $L = 128$, $T = 2400$ steps) the deviation is zero to machine precision, $\max_t |H - \ln K| = 0$. The observational entropy [17, 18] of the coarse observer decomposes as

$$S_{\text{obs}}(t) = H + I(t), \quad I(t) \geq 0, \quad (4)$$

where $I(t)$ is the information about the microstate hidden from the coarse observer (Appendix B). Since H is constant, *all* growth of S_{obs} is growth of I : in the runs of Fig. 1, 100% of ΔS_{obs} is carried by the hidden-information term, as an identity rather than a correlation. On the same runs the planted record decays from 1.000 bit to 0.006 bit. The record arrow is thus an exact information ledger: what the coarse observer loses about the past fact is part of what the coarse description hides about the microstate. This upgrades the familiar qualitative statement (record fidelity mirrors the entropy curve) to bookkeeping with no free play, and frames the quantitative question of this paper: *at what time is the ledger’s record entry effectively erased?*

T7: the record arrow as an exact information ledger (observational entropy) + the accessible record in bits

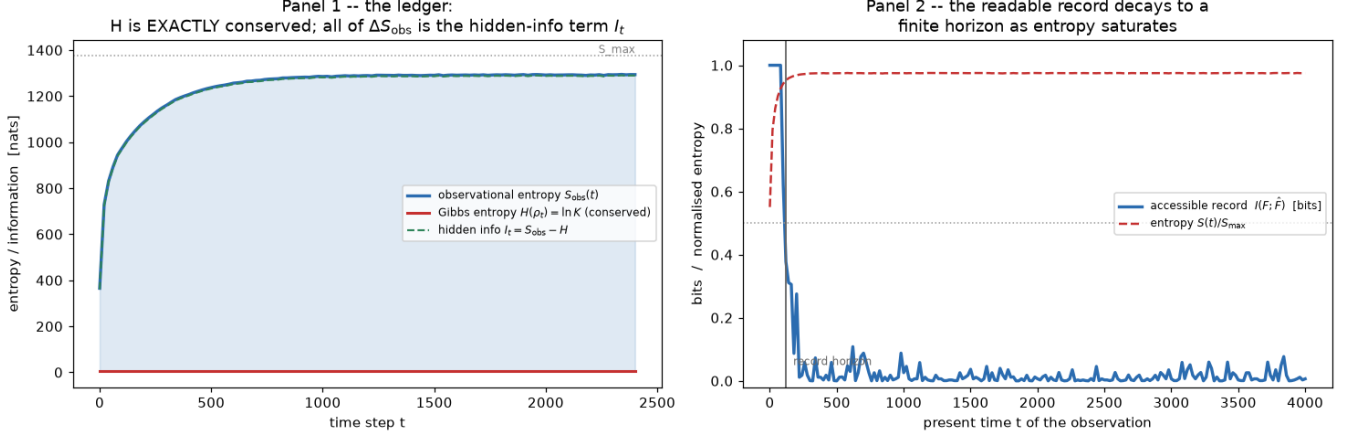


FIG. 1. **The record arrow as an exact information ledger** (bit-reversible lattice gas, $K = 64$ worlds, $L = 128$, $b = 8$). Left: the fine-grained entropy H of the evolving microstate ensemble is conserved to machine precision while the observational entropy S_{obs} rises; the rise is identically the hidden-information term I of Eq. (4). Right: on the same runs, the accessible record $I(F; \hat{F})$ of a planted one-bit fact decays from 1.000 to 0.006 bit as the gas thermalizes.

III. THE HORIZON LAW

Figure 2 shows the central measurement. We sweep the scrambling rate (scatterer fraction 0.15–0.42) with per-condition run lengths chosen so that no condition is clamped by the simulation window ($T = 400$ –2400 steps, sampling stride 2–6), five quenched-disorder seeds per condition, $K = 40$ worlds per fact class, $L = 64$, $b = 4$. Of 25 runs, 24 resolve both clocks (one run at the slowest condition remains censored and is excluded).

The result is a proportionality through the origin,

$$t^* = 1.08 t_S, \quad R^2 = 0.82, \quad (5)$$

over a $7\times$ range of t_S (26–192 steps). More important than the fit is its *flatness*: the per-condition ratios are 1.37, 0.97, 1.09, 1.02, 1.15 (standard deviations 0.21–0.28), so the law holds run by run rather than through one leverage point. (The largest ratio occurs at the fastest condition, where both times are shortest and the discrete sampling is coarsest relative to them.) The law is also robust in system size: repeating the measurement with the box and marker scaled together for $L = 48$ –128 (t_S spanning $6\times$) gives $\kappa(L) = 0.95, 0.89, 0.91, 1.23$, mean 0.99, with no systematic drift [26].

That κ is slightly above unity says only that the $\frac{1}{2}$ -bit record threshold is crossed slightly after the $0.9 S_{\text{max}}$ entropy threshold—a statement about conventions, not physics. The physical content, made explicit next, is that both observables relax on *one* clock.

IV. MECHANISM: ONE SLOW MODE, TWO THRESHOLDS

Why should the two clocks agree? In a finite ergodic system the late-time approach to equilibrium is dominated by the slowest relaxation mode; for our diffusive lattice gas this is the largest-wavelength density mode, with lifetime $\tau \sim L^2/D$. Both observables read that mode. The entropy deficit

$$\mathcal{D}(t) \equiv S_{\text{eq}} - S(t) \approx A_S e^{-t/\tau_S} \quad (6)$$

measures the large-scale inhomogeneity still to be erased, while the record signal

$$M(t) \equiv I(F; \hat{F})(t) \approx A_M e^{-t/\tau_M} \quad (7)$$

is the odd (left–right) large-wavelength component of the same relaxation: the planted fact is a parity of the slowest mode. If one timescale underlies both, $\tau_M \simeq \tau_S \equiv \tau$, the threshold crossings of Eqs. (2) and (3) give

$$t_S = \tau \ln \frac{A_S}{\theta_S}, \quad t^* = \tau_M \ln \frac{A_M}{\theta_M}, \quad (8)$$

with $\theta_S = 0.1 S_{\text{max}} - \mathcal{D}_{\text{eq}}$ (the deficit value at the entropy gate, $\mathcal{D}_{\text{eq}}$ the finite-size equilibrium deficit) and $\theta_M = \frac{1}{2}$ bit. Hence

$$t^* = \kappa t_S, \quad \kappa = \frac{\tau_M}{\tau_S} \frac{\ln(A_M/\theta_M)}{\ln(A_S/\theta_S)} \quad (9)$$

—a *computable ratio of logarithms*. Equation (9) explains the entire phenomenology at once, with no free parameter: (i) the proportionality itself; (ii) the flatness of κ

T7-horizon: the readable record dies exactly when entropy saturates — $t^* \approx t_S$ across a wide range of thermalization times

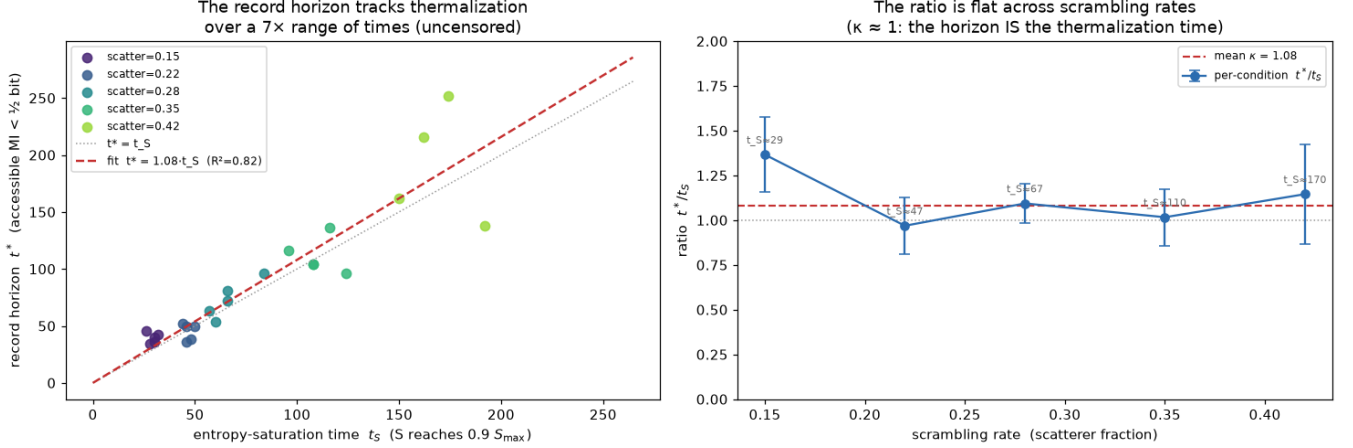


FIG. 2. **The record horizon tracks thermalization** (lattice gas; 5 scrambling rates $\times$ 5 disorder seeds; 24/25 runs resolved, nothing clamped). Left: t^* against t_S for all resolved runs; the through-origin fit gives $t^* = 1.08 t_S$ ($R^2 = 0.82$) over a $7\times$ range. Right: the per-condition ratio t^*/t_S is flat across scrambling rates—the law holds run by run.

in system size, because A_S scales with $S_{\max}$ so the logarithm ratio is size-free; (iii) the substrate dependence of κ (below: CA 1.08, hard disks 1.00, Clifford 0.79), because amplitudes and threshold conventions differ between substrates but enter only through slowly varying logarithms, keeping κ order one; (iv) the weak (logarithmic) sensitivity to threshold choices.

Figure 3 tests the mechanism directly (5 scrambling rates $\times$ 3 seeds; both tails fitted per run over their first decaying window, Appendix B). The tails are exponential with a *shared* timescale: $\tau_M/\tau_S = 0.92 \pm 0.30$ across all 15 runs, flat while the absolute τ_S varies by $7.6\times$ (11–84 steps; condition means 0.76–1.16 with no trend). The parameter-free postdiction of the crossing times from each run’s fitted (τ, A) and the thresholds lands at median error 5% for t^* and 1% for t_S ; the through-origin κ is postdicted at 0.98 against a measured 1.02 on the same runs. We conclude that the horizon law is a one-slow-mode theorem, not a numerical coincidence: *a record survives exactly as long as the slowest mode it is written in—and the entropy’s own approach to saturation is that same mode’s decay.*

V. UNIVERSALITY

A fair objection is that Secs. III–IV might describe an artifact of one substrate—a square lattice gas with quenched scatterers. We therefore re-ran the same two clocks in two deliberately different substrates. The target is *not* to reproduce the number $\kappa = 1.08$, which Eq. (9) says is convention-bound, but the invariant: $t^* \propto t_S$ with order-one flat κ , and a collapse of record decay and entropy relaxation onto one clock under t/t_S rescaling.

A. Continuous hard-disk gas

An event-driven hard-disk gas thermalizes by ballistic crossing rather than lattice scattering. The clean knob is a *self-similar* size sweep: the low-entropy blob (and the planted fact) scale with the box, so record and entropy live on the same spatial scale while the one thermalization time stretches with size. (Growing the box at fixed blob decouples two scales—filling the box versus erasing a local asymmetry—and the ratio then drifts; the self-similar sweep is the physically matched comparison.) Result (Fig. 4): $t^* = 1.00 t_S$ over a $5.1\times$ range ($t_S = 14$ –73), per-size $\kappa = 1.19, 1.03, 0.94, 0.97, 1.03$, and both observables collapse onto master curves under t/t_S . (This substrate is float-chaotic, so exact values vary at the percent level with the floating-point environment; an independent re-run on different hardware gives $\kappa = 1.04$ with the same flatness [26].)

B. Reversible quantum circuits

The bookkeeping of Sec. II is natively quantum [13, 17, 20]; the lattice gas exercises only its classical shadow. A Clifford brickwork circuit is unitary—hence exactly reversible—yet efficiently simulable [23, 24], allowing $N = 128$ qubits. Two clocks on the same random circuit: t_S is the half-chain entanglement saturation time (entanglement first reaches 0.9 of its late-time Page plateau [27, 28]); t^* is the time at which a reference qubit’s recoverable record $I(R : \text{local window})$ (window = $N/3$ sites, initially 2 bits for a maximally entangled reference) first drops below 1 bit. Sweeping $N = 24$ –128 stretches the ballistic scrambling time. Result (Fig. 5): $t^* = 0.79 t_S$ with $R^2 = 0.95$ over a $6.9\times$ range ($t_S = 33$ –229 layers),

T7-mechanism: $t^* \approx t_S$ because record and entropy relax through the SAME slowest mode — κ is a computable ratio of logs, not a fitted number

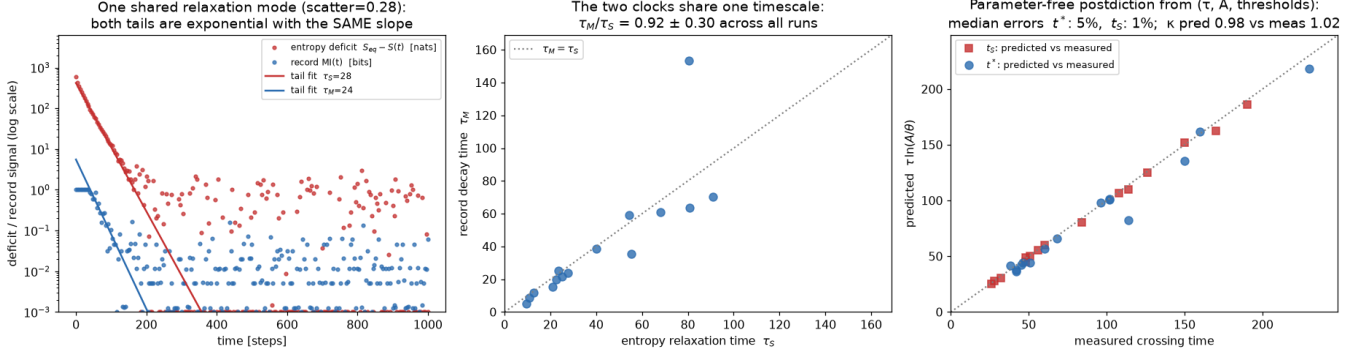


FIG. 3. **Why $t^* \approx t_S$: one slow mode, two thresholds** (lattice gas, 5 scrambling rates $\times$ 3 seeds). Left: for one example condition, the entropy deficit $S_{eq} - S(t)$ and the record signal $I(F; \hat{F})(t)$ on a logarithmic scale; both tails are exponential with the same slope. Middle: fitted τ_M against τ_S for all runs; the ratio is 0.92 ± 0.30 , flat over a $7.6\times$ range of absolute timescales. Right: parameter-free postdiction of both crossing times from the fitted (τ, A) and the thresholds, Eq. (9): median error 5% (t^*) and 1% (t_S); κ postdicted 0.98 vs. measured 1.02.

T7-md-horizon (universality axis 1): $t^* \approx t_S$ in a CONTINUOUS hard-disk gas — the record's lifetime is the thermalization time, not a lattice artifact

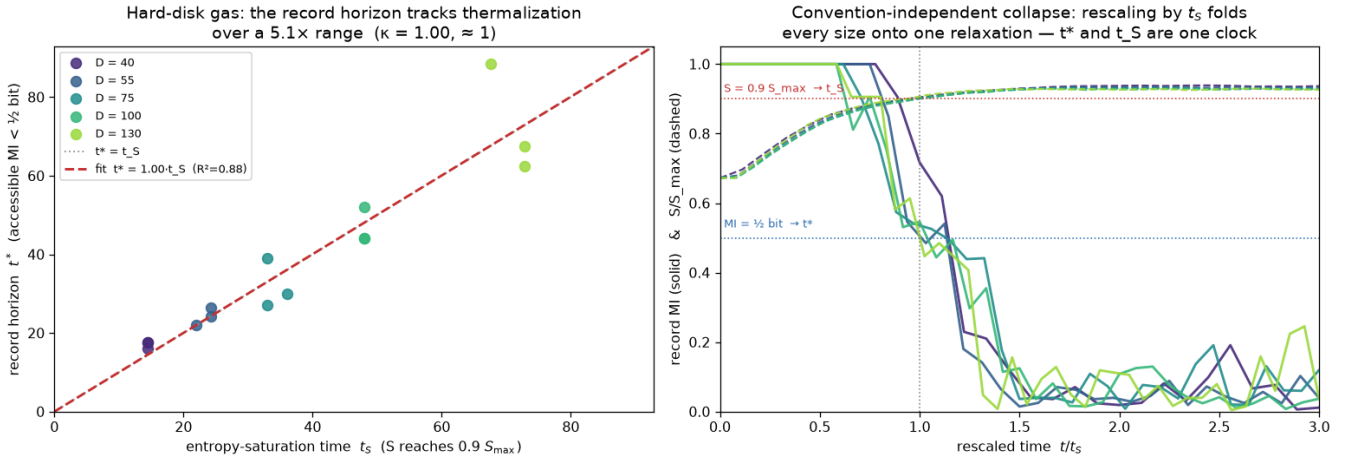


FIG. 4. **Universality I: continuous hard-disk gas** (self-similar size sweep, $D = 40-130$, three seed-ensembles per size). Left: t^* vs. t_S ; $t^* = 1.00 t_S$ over a $5.1\times$ range with flat per-size ratios. Right: rescaling time by t_S collapses the record MI (solid) and the entropy relaxation (dashed) of every size onto single master curves: t^* and t_S are one clock; κ is merely where each observable crosses its conventional threshold.

per-size ratios flat at 0.77–0.79 for $N \geq 64$, with the same t/t_S collapse.

Clifford circuits are simulable precisely because they are non-generic, so we add a non-Clifford control: a full statevector simulation at $N = 8-16$ with true von Neumann entropies, run under random Clifford gates ($\kappa = 0.89 \pm 0.32$) and genuinely universal Haar-random gates ($\kappa = 0.60 \pm 0.10$). Both obey the proportional law with order-one κ ; at these small sizes the absolute value is finite-size-suppressed, and the decisive point is that a universal gate set does not break the law. Had the horizon law been a stabilizer artifact, the Haar ensemble would have revealed it.

TABLE I. The horizon law across substrates. κ is convention- and substrate-dependent (Eq. (9)); the law is the flat proportionality.

Substrate	Dynamics	Range in t_S	κ
Margolus CA	bit-reversible, diffusive	$7\times$	1.08
size sweep	$L = 48-128$	$6\times$	0.99 (mean)
Hard-disk gas	Hamiltonian, ballistic	$5.1\times$	1.00
Clifford circuit	unitary, $N \leq 128$	$6.9\times$	0.79
Haar statevector	unitary, $N \leq 16$	(spot check)	0.60 ± 0.10

T7-clifford-horizon (universality axis 2 — quantum): $t^* \approx t_S$ in an exactly reversible Clifford circuit — the record's lifetime is the thermalization time

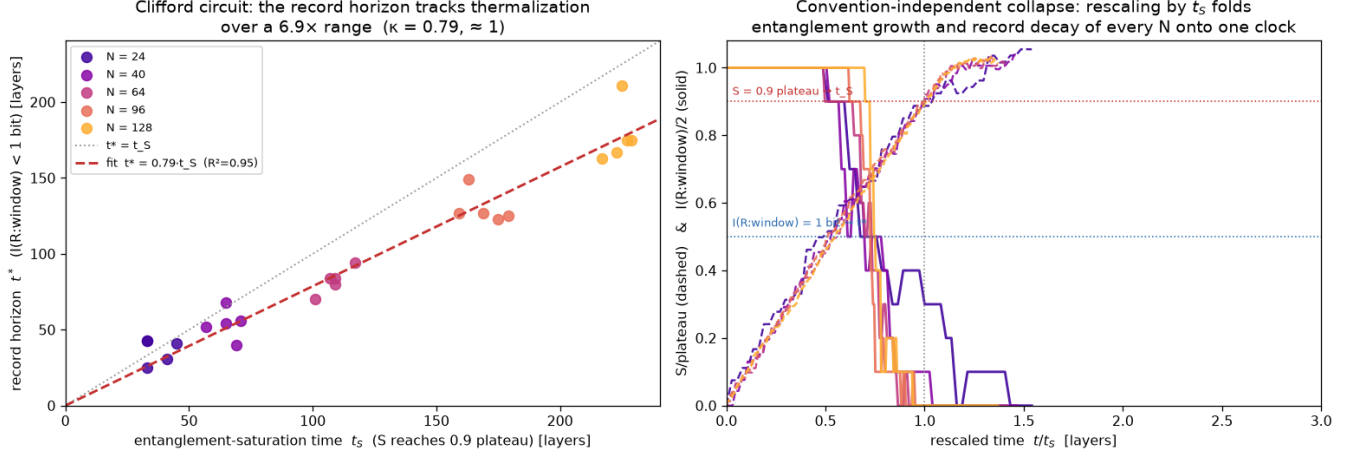


FIG. 5. **Universality II: exactly reversible quantum (Clifford) circuits** ($N = 24$ – 128 qubits, five circuit seeds per size). Left: record horizon vs. entanglement-saturation time; $t^* = 0.79 t_S$ ($R^2 = 0.95$) over a $6.9\times$ range, per-size ratios flat. Right: t/t_S collapse of entanglement growth (dashed) and record decay (solid) for all sizes. A statevector control with Haar-random gates (not shown; see text) confirms the law is not a stabilizer artifact.

VI. FALSIFIABILITY: A CONSERVATION LAW DECOUPLES THE CLOCKS

A law that cannot fail is a definition. The mechanism of Sec. IV says the horizon law should hold for *ergodic* reversible dynamics—the record must be written in a relaxing mode. This predicts its own failure mode: protect the record with a conservation law, and the two clocks must decouple.

We engineer this in the Clifford substrate (Fig. 6). A reference qubit is entangled with a record qubit at the chain end. The bulk scrambles with a random brickwork (so t_S stays finite), but the bond touching the record qubit is, with probability $1 - p_{\text{break}}$, a protected gate that uses the record only as a control, conserving Z_0 while still broadcasting it; with probability p_{break} it is a full scrambling gate. Thus p_{break} tunes ergodicity at the record. Result ($N = 64$, $5N$ layers, four seeds): t_S stays flat across the sweep (≈ 110 layers, relative std 0.03), while

$$\kappa(p_{\text{break}}) = 0.82, 0.74, 0.80, 0.88, 1.09, \geq 1.67, \geq 2.90 \quad (10)$$

for $p_{\text{break}} = 1, 0.5, 0.25, 0.1, 0.05, 0.02, 0$; at $p_{\text{break}} \leq 0.02$ the record outlives every simulated window (censored—the values quoted are lower bounds), and at $p_{\text{break}} = 0$ it is never lost. The horizon law is therefore *contingent on ergodic scrambling*: a conserved quantity is exactly what lets a record outlast—or point against—the entropy gradient. This is the extreme limit of a protected (pointer/code) subspace [12]; engineered quantum memory is, in this language, the deliberate construction of horizon-law violations. Mapping this boundary is what makes $t^* \approx t_S$ a falsifiable claim about generic dynamics rather than a tautology of our definitions.

VII. REDUNDANCY: CLASSICAL DARWINISM AND ITS SCALING

How is the record stored while it lives? Following the quantum Darwinism program [13, 15, 16] we measure the partial-information profile of the classical record: the mutual information I_f recovered by decoders restricted to random fractions f of the coarse cells, and the redundancy $R_\delta = 1/f_\delta$, where f_δ is the smallest fraction achieving $(1 - \delta)I_{\text{max}}$ (we use $\delta = 0.1$). Near the boundary the profile has the classical plateau shape: 10% of the cells already recover 89% of the record, and $R \approx 8$ – 10 independent copies coexist, collapsing to $R = 1$ at the horizon [26]—the classical analogue of the rise and fall of redundancy [16].

A. Scaling with environment size

A diffusion/signal-to-noise argument fixes the ideal scaling. Growing the system self-similarly (box L up at fixed cell size b , marker scaled with L) adds *equivalent* fragments: each added cell is an equally informative, conditionally independent detector of the planted parity near the boundary, so ideally one new copy per added fragment, $R \propto N$ with $N = (L/b)^2$, i.e. $\alpha = 1$ (ideal classical Darwinism). Refining the grid instead (b down at fixed L) multiplies fragments while dividing the per-fragment signal, so $\alpha < 1$ is expected—an SNR-confounded control.

Figure 7 shows the measured scaling over $N = 144$ to $16,384$ ($L = 48$ – 512 , six disorder seeds): R climbs from 7.3 to 438 . A power law over the full range gives $\alpha = 0.88$; the finite-size-free tail ($L \geq 128$) gives $\alpha = 0.91 \pm 0.05$;

T7-clifford-falsification: $t^* \approx t_S$ is contingent on ergodicity — a conserved quantity lets a record outlive thermalization ($t^* \gg t_S$)

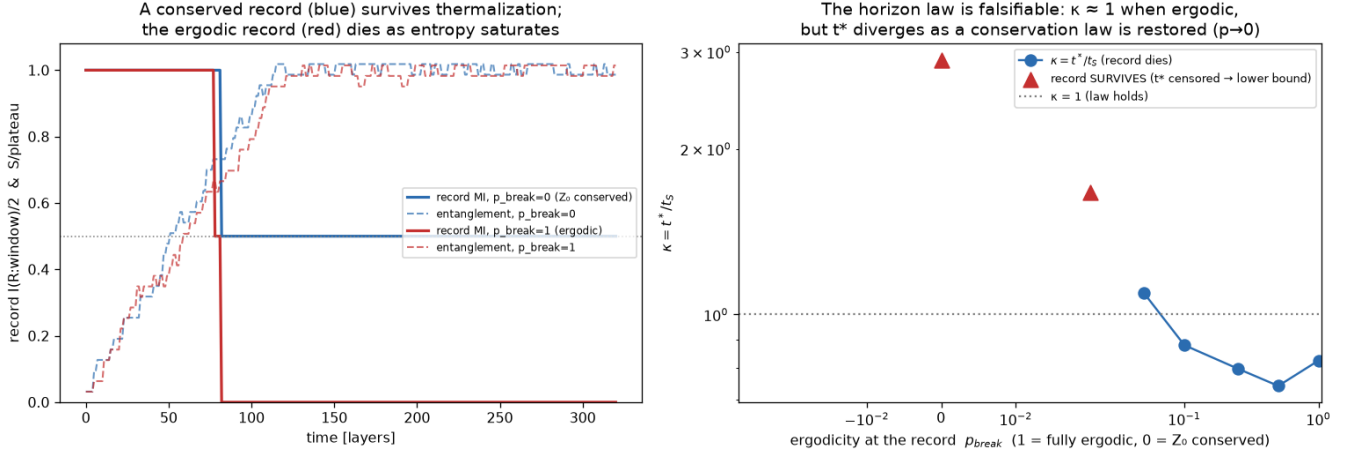


FIG. 6. **The law is falsifiable, and fails where it must** (Clifford chain, $N = 64$; a tunable Z_0 -conserving gate protects the record with probability $1 - p_{\text{break}}$). Left: example trajectories—under full scrambling the record dies as entanglement saturates, while the protected record survives arbitrarily long although the bulk thermalizes identically. Right: $\kappa = t^*/t_S$ against p_{break} ; the ratio is order one in the ergodic regime and diverges (censored points are lower bounds) as the conservation law is restored, while t_S stays flat.

and two independent extrapolations (a cutoff sweep in N_{min} and a correction fit $R = AN^\alpha(1 + c/N)$, mutually agreeing) give

$$\alpha_\infty = 0.92 \pm 0.04, \quad (11)$$

2.2σ below the ideal value—ideal Darwinism *supported but not certified* at accessible sizes. This quote is deliberately conservative: the data cannot distinguish the form of the finite-size correction (near-identical residuals), and the physically motivated $c/\sqrt{N}$ correction (the marker spans $\propto L = b\sqrt{N}$ cells) gives $\alpha_\infty = 0.95 \pm 0.07$, while $cN^{-1/4}$ gives 1.00 ± 0.12 (Appendix D). Meanwhile the SNR-confounded refinement sweep stays at $\alpha = 0.71$: the two ways of growing the environment separate cleanly, exactly as the argument predicts.

B. The quantum limit isolates the shortfall

The residual classical gap to $\alpha = 1$ is explained by its noiseless limit. In the stabilizer substrate we compare Zurek’s dichotomy directly (Fig. 8): a pointer bit *broadcast* into N_E environment qubits (decoherence-type dynamics) yields a flat partial-information plateau—any fragment carries the full classical record—and $R_\delta = N_E$ exactly across $N_E = 8$ –128, i.e. $\alpha = 1.000$: ideal Darwinism, realized. The same pointer under *scrambling* dynamics is encoded rather than broadcast: the partial-information profile becomes a step at half the environment and $R \approx 1$. Perfect discrete copying has no SNR loss, so it reaches the exponent the diffusive classical lattice only approaches; the classical sub-linearity is thereby

isolated as a signal-to-noise/finite-size artifact, not a limit of the redundancy principle. (We note honestly that broadcast redundancy is fragile: a few scrambling layers collapse R toward 1 well before the entropy horizon—redundant, multi-observer accessibility dies earlier than the single-copy record of Fig. 5, consistent with [16].)

VIII. DISCUSSION

A. What has been established

Within exactly reversible model dynamics prepared at a low-entropy boundary: (i) the growth of coarse entropy is identically the growth of information hidden from the coarse observer, with the fine-grained entropy conserved to machine precision—and a record’s decay is that same ledger read from the fact’s side; (ii) the readable lifetime of a record is the thermalization time, $t^* = \kappa t_S$ with order-one κ , flat across dynamics rates and sizes and reproduced across three substrates (Table I); (iii) the law has a mechanism—both clocks are slaved to the slowest relaxation mode, making κ a ratio of logarithms, postdicted parameter-free to a few percent; (iv) the law is falsifiable and fails precisely when a conservation law protects the record; (v) records are stored redundantly, with a scaling that approaches ideal classical Darwinism and attains it in the noiseless quantum broadcast.

Taken together, these results sharpen the qualitative slogan of the memory arrow [6–9]. Mlodinow and Brun argued that a robust record-forming system must sit on the entropy-increasing side of the gradient [8]; our measurements add the quantitative structure of that record-

T7-redundancy-scaling (high-resolution retry): pushing L to 512 ($N=16384$) resolves the finite-size-limited α toward ideal Darwinism

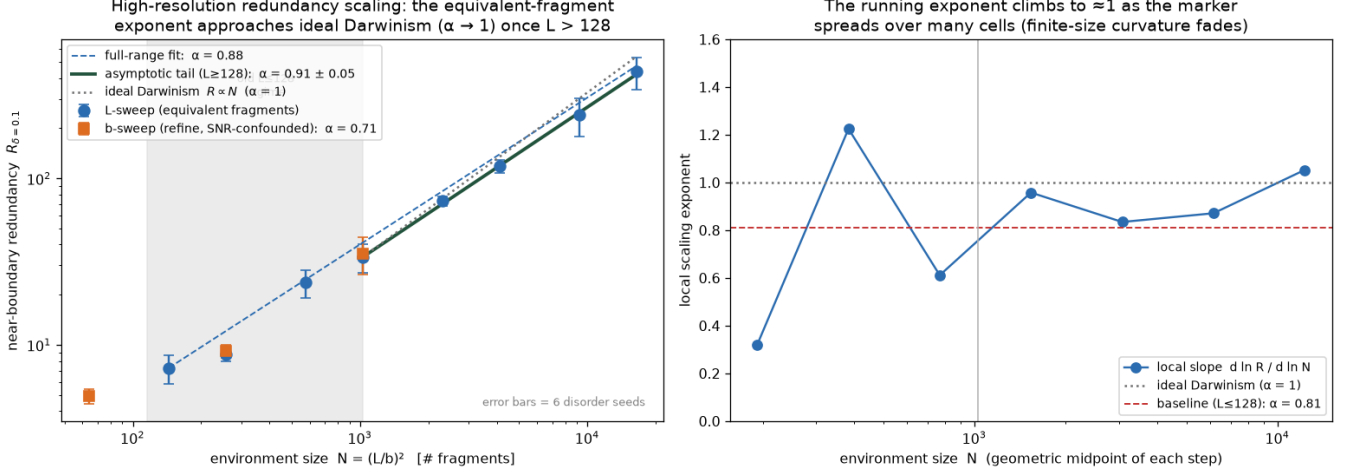


FIG. 7. **Redundancy scaling toward ideal Darwinism** (lattice gas, equivalent-fragment sweep $L = 48$ –512, i.e. $N = (L/b)^2$ up to 16,384; six disorder seeds). Left: near-boundary redundancy $R_{\delta=0.1}$ vs. N ; the full-range exponent is 0.88, the $L \geq 128$ tail 0.91 ± 0.05 (extrapolating to $\alpha_\infty = 0.92 \pm 0.04$, conservatively; see text), while the SNR-confounded refinement sweep stays at $\alpha = 0.71$. Right: the local slope $d \ln R / d \ln N$ climbs to ≈ 1 at the largest sizes as the small- L curvature fades.

T7-clifford-darwinism: the record is redundant under decoherence, encoded under scrambling; perfect broadcast reaches ideal Darwinism $\alpha = 1$

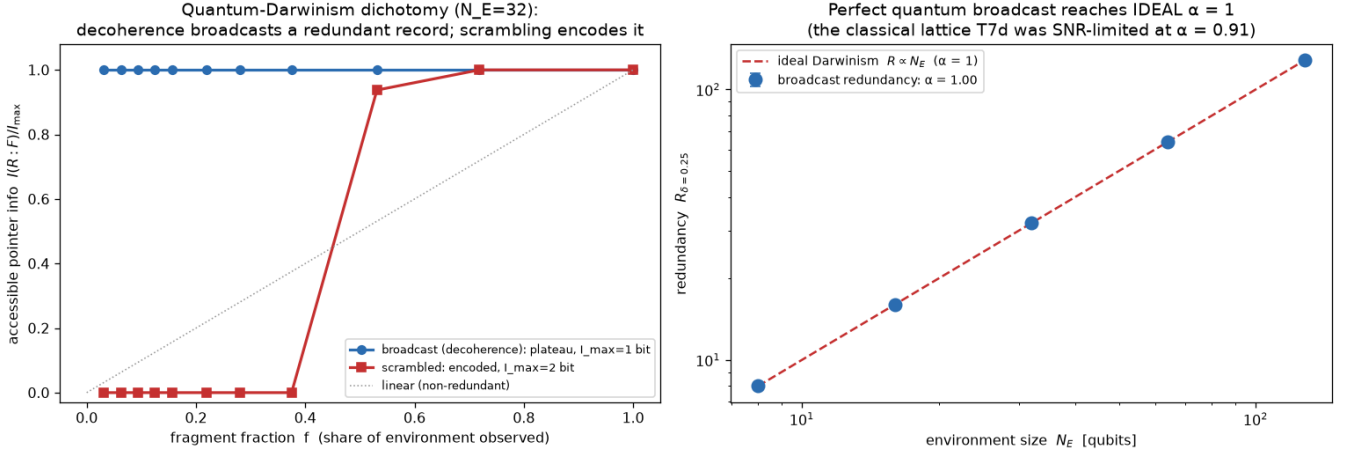


FIG. 8. **Quantum Darwinism in the reversible circuit substrate**. Left: partial-information profiles for a pointer bit broadcast into the environment (flat plateau: any small fragment carries the record) versus scrambled into it (step at half the environment: the record is encoded, not redundant). Right: redundancy of the broadcast record scales as $R_\delta = N_E$ exactly ($\alpha = 1.000$, $N_E = 8$ –128)—the ideal limit the classical lattice approaches from below.

keeping: a record is a slow mode of the very relaxation that defines the thermodynamic clock, so its readable lifetime is not a new parameter of nature but the thermalization time itself; its objectivity while alive is redundancy; and the only way to out-live the gradient is a conservation law—which is engineering (memory protection), not generic dynamics. In companion experiments in the same testbed [26], the *direction* of record accumulation flips with the sign of the entropy gradient (including for facts injected mid-run, realizing Reichen-

bach’s fork asymmetry), completing the picture: direction, lifetime, and storage of memory are all read off the gradient.

B. What has not been shown

First and fundamentally, nothing here derives the arrow of time: every run assumes the low-entropy boundary. This is not a defect of the testbed but a theorem-

level wall for any time-symmetric dynamics [4, 25]; our exactly reversible substrate makes it vivid, since entropy grows in *both* time directions away from the boundary state [26]. The mystery of the Past Hypothesis is used, not dissolved. (An exploratory companion experiment relocates the posit in the spirit of Carroll and Chen [29]—a *small* early volume plus expansion, rather than a fine-tuned microstate, sustains an unbounded two-headed entropy valley from a single bounce microstate evolved in both time directions—but the relocated question, why the volume was small, remains as open as the original [26].)

Second, κ 's numerical value is convention-laden; only the proportionality, the collapse, and the mechanism are physics. Third, the classical redundancy exponent is supported at, but not certified to, the ideal value ($\alpha_\infty = 0.92 \pm 0.04$ conservatively, up to 1.00 under equally consistent extrapolation forms); certification would require another decade in N . Fourth, all claims are about the model classes simulated here—though three substrates, a mechanism, and a mapped failure boundary argue that the horizon law is a generic property of ergodic reversible information dynamics rather than an artifact of any one model.

C. Outlook

Three extensions seem most valuable. (i) A quantitative tie of the shared timescale τ to transport: the mechanism predicts $\tau \propto L^2/D$ in the diffusive substrate, with D set by the scatterer density, so the horizon law should be expressible entirely in transport coefficients. (ii) A systematic threshold-sensitivity study, which the mechanism predicts to be logarithmic. (iii) In the corroboration direction [26]: conditioning equilibrium fluctuations on reproducing one half of a record and testing whether the other half corroborates would quantify why multiple independent records justify belief in a real low-entropy past—Reichenbach's common cause as a measurable statement.

ACKNOWLEDGMENTS

All simulations, per-experiment numerical verdicts, and figure-generating code are openly available [26].

Appendix A: The bit-reversible substrate

The Margolus automaton partitions the $L \times L$ periodic grid into 2×2 blocks, with the partition offset alternating each step. Each block is updated by a bijective lookup table acting on its 16 possible occupancies; the composition of bijections is a bijection, so the global map is exactly invertible, and running the inverse tables in reverse order returns the initial bitstring exactly (verified bit-for-bit at

every commit [26]). A quenched fraction (the scrambling knob) of blocks applies a rotation rule; the remainder apply the identity-preserving collision rule. Initial low-entropy states confine all particles to a width- w blob at density ρ ($w = 40$, $\rho = 0.45$, $L = 64$, $b = 4$ for the horizon experiments), with microstates sampled microcanonically. The hard-disk companion is an event-driven elastic-disk gas in a square box with specular walls; between collisions the flight is exact, so the only irreversibility is floating-point.

Appendix B: Records, decoders, and tail fits

The planted fact F places a marker of identical particle number on the left or right of the blob, so the classes share every conserved quantity. For each class, K microstates are evolved through the same quenched disorder ($K = 40$ for the horizon sweep, 64 for the ledger). At each sample time the coarse occupancy vectors of half the worlds train a nearest-centroid readout $\hat{F}$; $I(F; \hat{F})$ is computed from the held-out confusion matrix. This makes t^* an *operational* horizon: the time past which no such decoder of the present coarse state beats a half-bit of information about the fact.

For the ledger, an ensemble of K equiprobable microstates gives $H = \ln(\#\text{distinct microstates}) = \ln K$ under bijective evolution, while the coarse observer assigns the observational entropy [17, 19]; the hidden-information term of Eq. (4) is the difference, computed exactly from the evolving ensemble.

For the mechanism experiment, the equilibrium deficit $\mathcal{D}_{\text{eq}}$ (mean of the last 15% of each run) is subtracted before fitting; both tails are fitted by ordinary least squares on $\ln y$ over the first decaying crossing of a fixed window ($[0.04, 0.5]$ of the initial deficit for $\mathcal{D}$; $[0.04, 0.6]$ bit for M), which excludes the estimation floor that follows the record's death.

Appendix C: Stabilizer methods

Clifford states are tracked by the binary check matrix of their stabilizer group (phases dropped; they do not affect entanglement). Gates act as column operations [23, 24]. The entanglement entropy of a region A is $S_A = \text{rank}_{\mathbb{F}_2}(G|_A) - |A|$ ebits [30], and mutual informations follow from $I(A:B) = S_A + S_B - S_{AB}$. Brickwork layers apply independent random two-qubit Cliffords (random single-qubit words interleaved with CNOTs in both directions) to alternating bonds; this drives contiguous regions to volume-law (Page) entanglement [27, 28]. The protected gate of Sec. VI uses the record qubit only as a control, conserving Z_0 . The statevector control applies exact 4×4 unitaries (Haar-random via QR of complex Gaussians, or random Clifford words) and computes von Neumann entropies from Schmidt spectra.

Appendix D: Extrapolation of the redundancy exponent

Two estimators of the asymptotic exponent agree: (A) a cutoff sweep, fitting $\alpha_{\text{eff}}(N_{\min})$ on $N \geq N_{\min}$ and extrapolating linearly in $1/N_{\min} \rightarrow 0$ (giving 0.918); and (B) a correction fit $\ln R = c_0 + \alpha_{\infty} \ln N + c_1 N^{-p}$

with seed-bootstrapped errors. For $p = 1$: $\alpha_{\infty} = 0.922 \pm 0.036$; $p = \frac{1}{2}$: 0.947 ± 0.067 ; $p = \frac{1}{4}$: 1.001 ± 0.124 —with statistically indistinguishable residuals (RSS 0.060/0.063/0.064), so the main text quotes the most conservative value. The local slope $d \ln R / d \ln N$ reaches 0.96–1.05 across the largest size steps.

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